

CLASS XI · LECTURE 1

Information Technology and the Computer

Two periods · Start of Paper I · Uses and abuses of IT are a regular short question

By the end of this lecture you can

- Define computer, data, information, program and information technology.
- Draw the block diagram of a computer and name the four basic jobs.
- Write a short answer on uses or abuses of IT.
- Separate hardware from software with examples.

What a computer is

A computer is an electronic machine that accepts data, processes it according to a stored set of instructions, and produces a result. The raw facts and figures fed in are **data**. Data becomes **information** when processing gives it meaning. The set of instructions is a **program**, and programs together are **software**. The physical parts you can touch are **hardware**.

Information technology is the use of computers, storage and communication equipment to create, store, protect, exchange and use information. A college attendance system, a bank ATM and a WhatsApp message are all IT at work. The computer is the processing tool. IT is the whole activity of handling information with that tool and with networks.

DEFINITION Computer

An electronic device that takes data as input, processes it according to instructions, stores it when needed, and gives information as output.

Data, information and a program

Term	Meaning	Example
Data	Raw facts, not yet useful by themselves	75, 81, Ali, 12 May
Process	Work done on data by a program	Adding marks, sorting names
Information	Processed data that has meaning	Ali's total is 156 and he passed
Program	A set of instructions the computer follows	The mark-sheet program

The block diagram

Every computer, from a phone to a large server, does four jobs. Input brings data in. The processor works on it. Output presents the result. Storage keeps data and programs for later. Draw this whenever a question says “basic operations of a computer”.

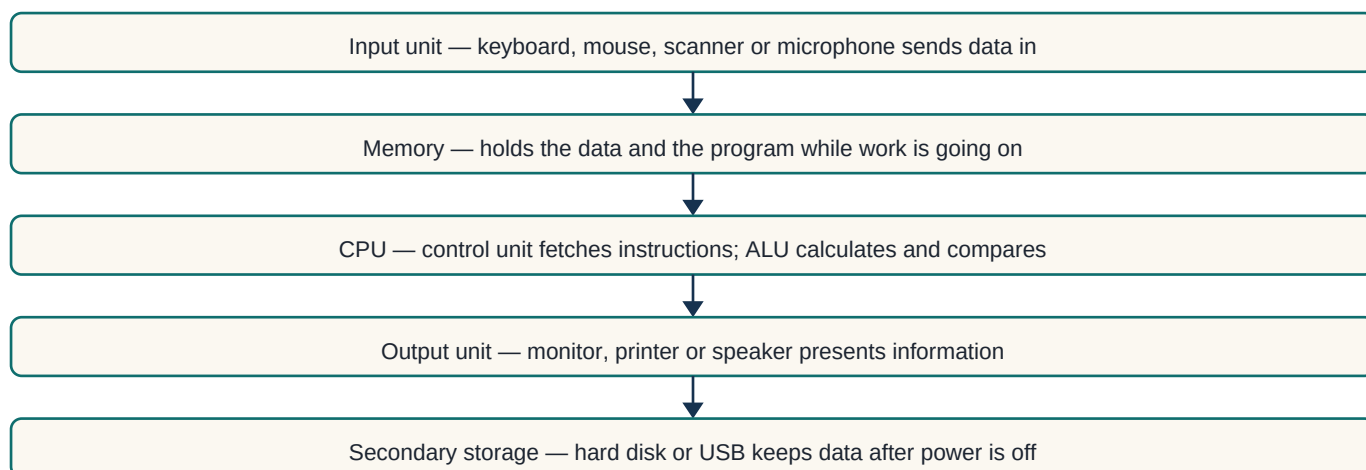


Figure 1.1 Block diagram of a computer system. Storage sits beside processing, not instead of it.

Why a computer is useful, and where it fails

Strength	What it means in an answer
Speed	Millions of operations in a second. A result that would take a person hours appears at once.
Accuracy	If the data and the program are correct, the result is correct. Errors come from bad input or a bad program. This is GIGO: garbage in, garbage out.
Diligence	A computer does not get tired or bored. The thousandth calculation is as careful as the first.
Storage	A disk holds a library of files and recalls any of them in a moment.
Versatility	The same machine can play a video, keep accounts, or run a lab program. The job changes when the program changes.

A computer has no feelings, no common sense and no judgement of its own. It cannot decide that a result is kind, fair or sensible unless a person has written that test into the program. It also needs electricity, and it can be attacked by harmful software.

Uses and abuses of IT

The short question is usually “What is information technology? Write any three uses or abuses.” Define IT in two lines, then give three uses **or** three abuses, each with a short example. Do not mix a long list of both unless the question asks for both.

Uses you can quote	Abuses you can quote
Education: lectures, online classes, exam results	Viruses and hacking that destroy or steal data
Banks: ATMs, transfers, statements	Software piracy: copying programs without a licence
Hospitals: records, scans, monitoring	Loss of privacy when personal data is shared or leaked
Business: stock, payroll, billing	Cyber fraud, fake messages and scams
Communication: email, video calls	Health strain from long hours at a screen, and wasted time
Government: NADRA, weather, defence systems	Unemployment in jobs that machines now do, if people are not retrained

EXAM TIP Shape of a 3-mark answer

Line 1: definition of IT. Lines 2–4: three uses, each one clause plus an example. Stop. Five vague words such as “IT is used everywhere” do not score. Named examples do.

Hardware and software, first distinction

Hardware is the physical equipment: keyboard, mouse, monitor, printer, system unit, cables. If the power is off, the hardware is still there. Software is the set of programs that tell the hardware what to do. A printer is hardware. The program that sends a page to the printer is software. You cannot run hardware usefully without software, and software cannot run without hardware.

ON THE BOARD One sentence to memorise

Hardware is touched. Software is run. Data is entered. Information is the result.

Checkpoint

Try the questions before you read the model answers. Section B answers on the real paper should stay within about six or seven lines.

Multiple choice

1. Processed data that has a meaning is called:

- A) Hardware
- B) A program
- C) Information
- D) A bus

2. GIGO means:

- A) The CPU is slow
- B) Wrong input produces wrong output
- C) A type of printer
- D) A network cable

Short answers

S1. What is information technology? Write any three uses of IT.

S2. Write any three abuses of information technology.

Detailed answer

L1. Draw the block diagram of a computer and explain its basic operations.

Model answers

Cover this section while you attempt the checkpoint. The wording is a full-mark style answer, not the only acceptable wording.

MCQ 1. C. Data is raw. Information is data after processing.

MCQ 2. B. Garbage in, garbage out. The machine follows the data it is given.

S1. Information technology is the use of computers and communication equipment to create, store, exchange and use information. Three uses are: banks use IT for ATMs and money transfers; hospitals use it for patient records and scans; colleges use it for results, timetables and online classes.

S2. IT can be abused when a virus damages files, when a person copies software without a licence (piracy), and when private data is stolen or published. Cyber fraud, in which a fake message tricks a person into sending money, is a fourth common abuse.

L1. A computer performs input, processing, output and storage. The input unit (keyboard, mouse or scanner) accepts data and converts it into a form the computer can handle. Memory holds the program and the data during the work. The CPU has a control unit, which fetches and decodes instructions, and an ALU, which calculates and compares. The output unit (monitor or printer) presents information. Secondary storage, such as a hard disk, keeps programs and data after the power is switched off. Draw five labelled boxes: Input, Memory, CPU, Output and Storage, with arrows from Input to Memory and CPU, from CPU to Output, and both ways between CPU and Storage.